Feasibility Ceilings Before Learning: Sizing and Timing Bounds for Drone-Mesh Interception

Dmytro Babarytskyi
Kharkiv National University of Radio Electronics
Kharkiv, Ukraine
dmytro.babarytskyi@nure.ua
babaritskiydmytro@gmail.com
ORCID: 0009-0009-3614-9601

Abstract*—A mesh of interceptors defending a fixed site must spread over the predicted arrival point of an approaching object. This is a recurring task in multi-agent robotics: N agents, each covering a disc of radius r, spread over a probability density so that a point drawn from it falls within some agent's disc. We give closed forms for the quantities that bound performance before any policy is chosen, and validate each by Monte Carlo. The uncertainty of a point predicted by extrapolating a noisy linear track follows from least-squares variance propagation plus a delta-method term for the uncertain arrival time, agreeing with simulation to within 1.3%. The coverage probability of a packed mesh admits a closed form accurate to 0.007 absolute, with a volumetric analogue. A naive area ratio misses a ceiling: while agents stay at least $2r$ apart, no mesh exceeds 90.69% coverage in the plane or 74.05% in a volume. This ceiling belongs to the separation constraint, not to the problem — permitting overlap substitutes Kershner's thinnest-covering density and removes it, the better regime above a crossover at $p = 0.80$. Hexagonal packing is optimal among five arrangements; the centroidal-Voronoi configuration coverage control supplies reaches only half its coverage at small budgets. The Gaussian assumption is conservative under heavy tails and anisotropy but fragile under bias: a one-sigma aim-point offset costs 15 points of coverage. A repositioning team must also choose a commit time, and the admissible window can be empty. Against the applicable ceiling, analytic placement attains 85% of what the configuration permits while centralised PPO averages 15% over three trainer seeds, with a paired bootstrap against a null policy containing zero at every seed. We argue such ceilings should be computed before any learning experiment and quoted alongside every result.

***Index Terms*—Counter-UAS, interception, site protection, aerial robotics, multi-agent systems, coverage control, circle packing, uncertainty propagation, feasibility analysis, reinforcement learning, negative results, reproducibility.**

I. Introduction

A fixed site is approached by an airborne object on a ballistic or near-ballistic course. A sensor tracks the object and predicts where it will cross a defensive plane standing off from the site, with an error that shrinks as the track lengthens. A mesh of small interceptors, each able to engage whatever comes within a radius r of it, must spread itself across that predicted crossing point and be on station before the object arrives. Three questions decide whether the concept is viable at all, and none of them is about control: how many interceptors the prediction's uncertainty demands, how early the mesh must commit to a position, and whether any commit time satisfies both the sharpness of the prediction and the distance the vehicles must fly. All three can be answered from geometry, before a controller exists.

We strip the application to that geometry. Consider a team of N agents constrained to a plane, each able to cover a disc of radius r centred on itself. A point of interest will arrive somewhere in that plane at a known future time; the agents hold a noisy prediction of where. The team must distribute itself so that the arriving point falls inside some agent's disc. This is a coverage problem in the sense of Cortés *et al.* [1], specialised in two ways that turn out to matter a great deal. The density is a prediction with a quantified and time-varying error rather than a static importance map. And success is binary and instantaneous — a point either lands within r of some agent or it does not — rather than an integrated sensing cost.

Tasks of this shape invite a multi-agent reinforcement-learning treatment, and the recipe is easy to follow: define an environment, shape a reward, train a policy, report a success rate. The difficulty is that a bare success rate on a hand-designed task carries almost no information. Whether 16% is a failure or a near-optimal result depends on what the configuration permits, and that quantity is rarely computed. Henderson *et al.* [30], Colas *et al.* [31] and Agarwal *et al.* [32] have each documented how fragile deep reinforcement-learning comparisons are to precisely this kind of missing context — unreported baselines, too few seeds, and point estimates quoted without dispersion.

This paper supplies the missing context for the coverage task above, in closed form, and then validates the assumptions the closed form rests on. We contribute:

1) the uncertainty σ of the predicted arrival point, propagated from per-fix sensor noise through a least-squares track fit, with a delta-method term that a common heuristic omits at a cost of up to $9.5\times$ (Section IV);
2) the coverage probability attainable by N agents at given σ and r, in the plane and in a volume, including a ceiling no mesh can exceed at any size (Section V);

3) an empirical check that hexagonal packing is the right arrangement, and a quantification of how far the standard coverage-control answer falls short of it on this objective (Section VI);
4) a robustness study of the Gaussian assumption, separating departures that make the law conservative from the one that breaks it (Section VII);
5) the admissible window of commit times for a team that must physically reposition, which can be empty (Section VIII).

Section X reports what happened when a learned placement policy was trained on this task, across three trainer seeds, and argues that the outcome is better explained by these bounds than by any property of the learner. Section XI states what the analysis omits.

Each individual result applies standard material — ordinary least squares, the delta method, densest packing of equal spheres. The claim is that computing them *first*, and reporting them *alongside* learned results, changes what those results mean, and that this is not current practice.

II. Related Work

A. Coverage control

The canonical formulation distributes agents over a density to minimise an integrated sensing cost, and is solved by Lloyd-type descent to a centroidal Voronoi configuration [1], [2]. The framework has been extended to agents that learn the density online [3] and developed into a general theory of distributed robotic networks [4]. More recently, graph neural networks have been used to let agents exploit multi-hop information and beat the purely local classical controller [5].

The objective in all of this work is continuous: a weighted integral of the distance from each point to its nearest agent. Our objective is the probability that a single draw lands within a fixed radius of some agent. The two are not equivalent, and Section VI measures the gap: at small budgets, the centroidal Voronoi configuration attains barely half the coverage of a packed one, because minimising mean squared distance rewards spreading into the tails while binary containment does not.

B. Packing bounds

That the densest packing of equal circles in the plane has density $\pi/\sqrt{12}$ is classical, due to Thue and developed by Fejes Tóth [6], [7]; the volumetric analogue $\pi/\sqrt{18}$ is the Kepler conjecture, proved by Hales [9] and later verified formally by machine [10]. We use these as the ceiling on what any non-overlapping arrangement of covering discs can achieve — a qualification that matters, and which Section V-E makes explicit by showing what happens when the non-overlap constraint is dropped. To our knowledge this ceiling is not usually stated in the multi-agent coverage literature, where it is natural to write down an area ratio that tacitly assumes perfect tiling and therefore permits coverage probabilities that are unattainable at any team size.

C. Track prediction and manoeuvre models

Propagating measurement noise through a least-squares trajectory fit is standard estimation practice [12], as is the recursive alternative [11]. We use the batch closed form rather than a filter because the quantity wanted is not a running state estimate but the error covariance of an extrapolation to a fixed plane, and because a closed form keeps the resulting sizing law analytic end to end. The delta method used for the uncertain arrival time is likewise textbook [15].

The straight-line assumption is the weak point, and the tracking literature has long known it. Singer [13] models target acceleration as a correlated random process precisely because constant-velocity fits degrade badly under manoeuvre, and Li and Jilkov [14] survey several decades of successors. Section VIII uses a deliberately crude version of this idea — an independent per-axis acceleration with fixed standard deviation — and finds that it dominates the sensor term by two orders of magnitude at realistic lead times. A Singer-type correlated model would refine the constant; it would not change which term binds.

D. Assignment

Allocating agents to positions at minimum total travel is a linear assignment problem, solved exactly by the Hungarian method [16] in $O(N^3)$. Distributed alternatives exist where centralised computation is unavailable [17]–[19]. This matters below because assignment is one of two classical sub-problems that an end-to-end policy is implicitly asked to rediscover, and the only one with an exact polynomial solution.

E. Interception and pursuit

Coverage of a predicted arrival point is adjacent to pursuit-evasion, surveyed for mobile robotics by Chung *et al.* [20]. The distinction is that the team here does not pursue: it positions itself ahead of the arrival and lets the geometry do the rest, which converts a speed problem into a placement problem. That is also why packing, rather than pursuit strategy, is the governing mathematics.

F. Multi-agent reinforcement learning and its evaluation

PPO [21] remains a common default for continuous multi-agent control, and is a strong baseline in cooperative settings when implemented carefully [22]; we use the Stable-Baselines3 implementation [23]. Permutation-equivariant architectures [24]–[26] are the standard remedy for policies whose action space treats interchangeable agents as unrelated output dimensions. Our negative result in Section X concerns an unstructured centralised baseline, and we are explicit that it does not settle the structured case.

Reward-specification pathologies are well understood in principle [27]–[29]; what we add is a concrete instance in which two of them produced the same observable symptom as a genuine learning failure, and were diagnosed only after the fact. On evaluation, we follow the recommendations of [30]–[32] in reporting multiple seeds with interval estimates rather than point estimates, using the percentile bootstrap [33].

III. Problem Formulation

A point of interest travels on a straight line and will cross a fixed plane $x = x_p$ at an unknown in-plane location $\mathbf{c} \in \mathbb{R}^2$. A sensor supplies n position fixes at times $t_1, \ldots, t_n$, each corrupted by independent Gaussian noise of per-axis standard deviation σ_s.

N agents occupy the plane. Each covers a disc of radius r; r is treated throughout as a fixed input parameter of the platform, not a design variable. Agents move with bounded speed $v_{\max}$ and bounded acceleration a, and must be stationary on station when the crossing occurs. Success is the event that $\mathbf{c}$ lies within r of at least one agent, and the objective is its probability,

$$p \;=\; \Pr\Big[\min_{i\in\{1,\ldots,N\}} \|\mathbf{c} - \mathbf{q}_i\| \le r\Big], \tag{1}$$

over agent positions $\mathbf{q}_i$ subject to a minimum separation s.

The parameter values carried through the paper belong to the setting of Section I: $\sigma_s = 4$ m per axis is representative of a radar track on a small airborne object at the ranges involved, $v_{\max} = 33$ m/s and $a = 27$ m/s^2 are of the order of a small multirotor interceptor, and the manoeuvre magnitudes swept in Section VIII run from $0.1g$, a nearly ballistic object, to $3g$, one that is actively evading.

Two modelling choices need stating, because the results depend on them and neither is forced. First, r is a *coverage* radius, a property of whatever the agent senses or affects, whereas s is a *physical* constraint on how close two agents may fly. These are distinct quantities and need not coincide. Throughout Sections V to VIII we assume $s \ge 2r$, so that coverage discs are disjoint and the arrangement is a packing; Section V-E treats the alternative and shows it changes the conclusions materially. In the configuration of Section X, $s = 2.5$ m against $2r = 6$ m, so the packing regime there is a choice of arrangement rather than a constraint of the platform.

Second, N is treated as a continuous quantity wherever it is the output of a sizing formula. A requirement of "22.1 agents" means that 22 are insufficient and 23 suffice; we report the continuous value because rounding discards the margin, which matters when comparing designs whose requirements differ by less than one agent.

We write σ for the per-axis standard deviation of the predicted crossing point and $R = \sigma\sqrt{2}$ for its RMS radial error. Packing densities are $\eta_2 = \pi/\sqrt{12} \approx 0.9069$ in the plane and $\eta_3 = \pi/\sqrt{18} \approx 0.7405$ in a volume. T denotes the commit time, the lead with which the team fixes its aim point, and C the half-width of the corridor within which the crossing may fall.

All experiments import the prediction, lattice and environment code from the accompanying prototype rather than reimplementing it, so that a validation run tests the implementation that actually runs. Numerics use NumPy [34] and SciPy [35].

IV. Prediction Uncertainty

A. Closed form

Fit per-axis ordinary least-squares lines $p_k(t) = a_k + b_k t$ for $k \in \{x, y, z\}$ and solve $p_x(t) = x_p$ for the crossing time t_c. With $\bar{t}$ the mean fix time and $S = \sum_i (t_i - \bar{t})^2$, the standard OLS results are

$$\mathrm{Var}(b) = \frac{\sigma_s^2}{S}, \quad \mathrm{Var}(a) = \sigma_s^2\left(\frac{1}{n} + \frac{\bar{t}^2}{S}\right), \quad \mathrm{Cov}(a,b) = -\frac{\sigma_s^2 \bar{t}}{S}. \tag{2}$$

The fitted value at an arbitrary time τ therefore has variance

$$\mathrm{Var}\big(a + b\tau\big) = \mathrm{Var}(a) + \tau^2 \mathrm{Var}(b) + 2\tau\,\mathrm{Cov}(a,b). \tag{3}$$

The crossing time is itself an estimate. Applying the delta method [15] to $t_c = (x_p - a_x)/b_x$, with $\partial t_c/\partial a_x = -1/b_x$ and $\partial t_c/\partial b_x = -t_c/b_x$,

$$\mathrm{Var}(t_c) = \frac{\mathrm{Var}(a_x)}{b_x^2} + \frac{t_c^2}{b_x^2}\mathrm{Var}(b_x) + \frac{2t_c}{b_x^2}\mathrm{Cov}(a_x, b_x). \tag{4}$$

Each in-plane axis then carries its own fit variance at $t = t_c$ plus a term from t_c being uncertain, since to first order the fitted value shifts by $b_k\,\delta t_c$. Combining,

$$R^2 = \sum_{k\in\{y,z\}} \Big[\mathrm{Var}\big(a_k + b_k t_c\big) + b_k^2\,\mathrm{Var}(t_c)\Big]. \tag{5}$$

The second term is easy to omit and expensive to omit. Error in the fitted closing velocity does not merely blur the crossing *time*: it is multiplied by t_c before it reaches the in-plane prediction, so a small slope error compounds into a large position error at long lead. Two practical notes follow. The term vanishes identically for a point travelling exactly along the x axis, so a validation scenario built without lateral velocity will not exercise it. And because the sensor noise is identical and independent across axes in our setting, there is no cross-term between the x fit and the y, z fits to add.

B. Validation

Table I reports (5) against Monte Carlo over noisy redraws of a fixed true trajectory, on a 4×4 grid. The grid is parameterised by (n, t_c) rather than by $(n, \text{standoff})$: holding the standoff fixed while adding fixes consumes the whole approach and leaves nothing to extrapolate, which confounds the two effects under test. The standoff is solved for instead, so every cell is a valid geometry.

Agreement is within 1.3% in the worst cell. Calibration — the fraction of trials whose true crossing point falls inside the radius the predictor reported — holds between 62.1% and 64.7% against the 63.2% nominal for a one-sigma radial bound on a two-dimensional Gaussian. This is the stronger of the two checks: agreement in magnitude could be arranged by a lucky constant, whereas calibration across sixteen geometries tests the shape of the distribution as well.

For comparison the table carries a hand-tuned heuristic of the form $\sigma_s/\sqrt{n} + 0.15\,\sigma_s t_c$, which captures the direction of every effect and none of their sizes. It understates the true uncertainty by between $2.1\times$ and $9.5\times$, and the understatement

TABLE I
CLOSED-FORM RADIAL UNCERTAINTY VS. MONTE CARLO ($\sigma_s = 4$ M, 4 000 TRIALS PER CELL)

n	t_c (s)	R (m)	MC (m)	err. (%)	heur. (m)	calib. (%)
3	0.5	9.60	9.58	0.2	2.61	63.2
3	1.0	14.42	14.32	0.7	2.91	64.1
3	2.0	24.30	24.53	1.0	3.51	63.0
3	4.0	44.26	44.63	0.8	4.71	63.1
5	0.5	6.36	6.44	1.2	2.09	62.4
5	1.0	8.46	8.49	0.3	2.39	62.4
5	2.0	12.81	12.78	0.3	2.99	62.8
5	4.0	21.68	21.65	0.1	4.19	63.1
8	0.5	4.62	4.62	0.0	1.71	63.3
8	1.0	5.62	5.59	0.6	2.01	64.1
8	2.0	7.71	7.72	0.1	2.61	63.3
8	4.0	11.99	12.10	0.9	3.81	62.7
12	0.5	3.60	3.59	0.2	1.45	63.3
12	1.0	4.13	4.13	0.2	1.75	62.1
12	2.0	5.24	5.29	0.9	2.35	62.2
12	4.0	7.53	7.44	1.3	3.55	64.7

grows with t_c — which is the $b_k^2 \mathrm{Var}(t_c)$ term reasserting itself. Fig. 1 shows both comparisons.

V. COVERAGE LAWS

A. Why the area ratio is not enough

A natural way to size the mesh is an area ratio: to cover the disc of radius $m\sigma$ with non-overlapping discs of radius r, take $N \approx (m\sigma)^2/r^2$. This is optimistic, because it assumes the discs tile the plane perfectly. They cannot: the densest packing of equal circles covers only $\eta_2 = \pi/\sqrt{12}$ of the plane [6].

B. The planar law

Carrying the packing density through, N touching discs of radius r occupy a roughly circular region of area $N\pi r^2/\eta_2$, that is, out to radius $\rho = r\sqrt{N/\eta_2}$. Within that region a fraction η_2 of the area is covered, and the enclosed mass of an isotropic Gaussian is $1 - \exp(-\rho^2/2\sigma^2)$. Hence

$$p(N) = \eta_2 \left(1 - \exp\left(-\frac{Nr^2}{2\eta_2\sigma^2}\right)\right), \tag{6}$$

which inverts to

$$N(p) = -\frac{2\eta_2\sigma^2}{r^2}\,\ln\left(1 - \frac{p}{\eta_2}\right). \tag{7}$$

Two consequences follow immediately. The count scales with the *square* of σ/r, with only the constant depending on the target coverage. The exponent is exact by construction. Fitting it back from the closed form would therefore be circular; inverting Monte Carlo measurements instead — for each (σ, r), the smallest N whose *measured* coverage reaches a target — gives a fitted exponent of 1.876, with the counts themselves agreeing with (7) to within 16% in the worst case. The shortfall from 2 sits entirely at the smallest budgets, where an integer number of discs cannot track a continuous law. And $p(N) \to \eta_2$ as $N \to \infty$, so no arrangement of non-overlapping discs, however large, covers more than 90.69% of the density. The area-ratio rule, which permits $p \to 1$, lacks this ceiling and is therefore not merely imprecise but qualitatively wrong in the regime where it matters most.

TABLE II
THE AREA-RATIO RULE $N = (m\sigma)^2/r^2$, AUDITED ($\sigma = 9.25$ M, $r = 2$ M, 200 000 TRIALS PER CELL)

m	N by rule	claimed p	actual p	N required
1.0	22	0.394	0.388	22.1
1.5	49	0.675	0.643	53.0
2.0	86	0.865	0.806	119.0
2.5	134	0.956	0.877	unreachable

C. Validation and audit

Against Monte Carlo using the production hexagonal-lattice generator, (6) is accurate to 0.007 absolute across $N \in [1, 130]$ (Fig. 2a). Table II audits the area-ratio rule at $\sigma/r = 4.6$: what it claims, what it delivers, and what N would actually be required. At $m = 2$ it claims 86.5% and delivers 80.6%, and reaching the claimed value needs 119 agents rather than 86, an inflation of 1.38×. At $m = 2.5$ it claims 95.6%, a figure above the packing ceiling and therefore unattainable at any N.

D. The volumetric law

If the arrival altitude is also uncertain, the mesh must cover a volume. The derivation carries over with two substitutions: $\eta_3 = \pi/\sqrt{18}$ for the densest sphere packing [9], [10], and the chi distribution with three degrees of freedom for the enclosed mass,

$$F_3(\rho;\sigma) = \mathrm{erf}\left(\frac{\rho}{\sigma\sqrt{2}}\right) - \sqrt{\frac{2}{\pi}}\,\frac{\rho}{\sigma}\,\exp\left(-\frac{\rho^2}{2\sigma^2}\right), \tag{8}$$

with $\rho = r(N/\eta_3)^{1/3}$ and $p_3(N) = \eta_3 F_3(\rho;\sigma)$. Against Monte Carlo over a face-centred-cubic lattice this is accurate to 0.005 absolute (Fig. 3). The exponent on σ/r is exactly 3, again by construction rather than by measurement: ρ scales as $(N/\eta_3)^{1/3}$, so inverting at fixed p gives a cubic dependence identically. The empirical content here is the agreement of $p_3(N)$ with Monte Carlo, not the exponent. The volumetric ceiling, under the same separation assumption as before, is 74.05%.

The practical significance is the ratio. At $\sigma/r = 4.6$ and $p = 0.5$, a planar mesh needs 31 agents and a volumetric one 473 — a factor of 15. Constraining the arrival to a known plane is worth more than any plausible improvement in the agents themselves.

E. The ceiling belongs to the constraint, not to the problem

Everything above assumes $s \geq 2r$. That assumption is doing more work than it appears to, and it is worth isolating.

If the platform tolerates $s < 2r$, coverage discs may overlap and the arrangement is no longer a packing but a *covering*. The governing constant changes accordingly: the thinnest covering of the plane by equal circles has density $\theta_2 = 2\pi/\sqrt{27} \approx 1.2092$, established by Kershner [8]. Because $\theta_2 > 1$, overlap

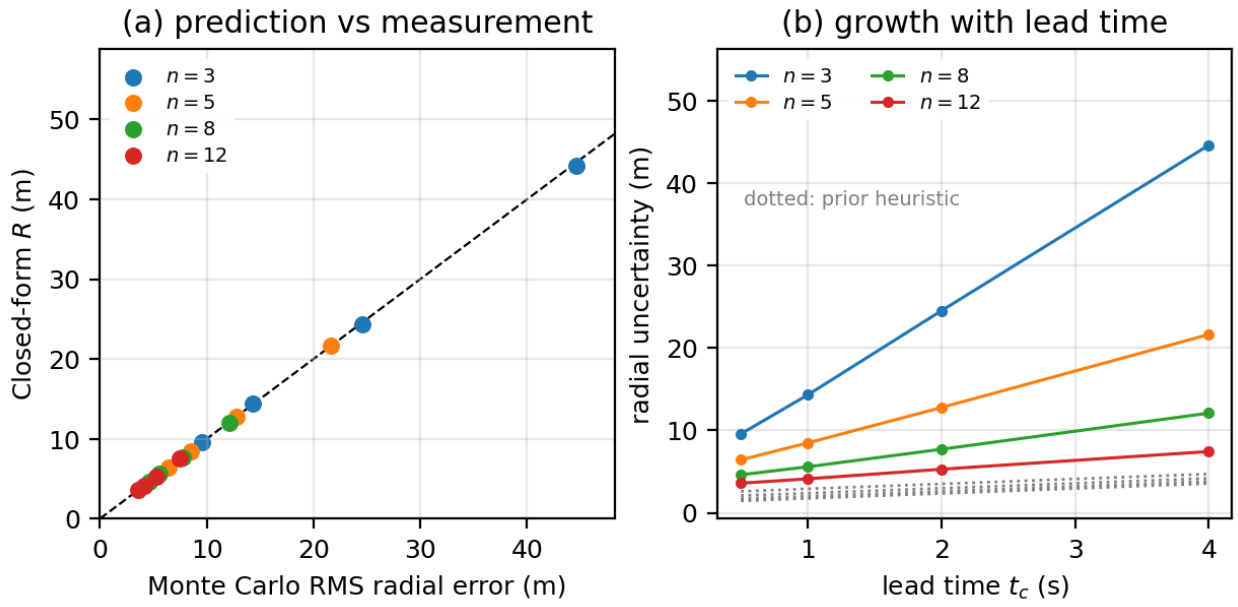


Fig. 1. (a) Closed-form R against Monte Carlo RMS radial error across the grid of Table I; the dashed line is $y = x$. (b) Growth of the uncertainty with lead time t_c; dotted curves are the hand-tuned heuristic, whose error grows with t_c.

wastes area — N discs now blanket a region of area only $N\pi r^2/\theta_2$ — but the blanket has no gaps, so

$$p_{\text{cov}}(N) = 1 - \exp\left(-\frac{Nr^2}{2\theta_2\sigma^2}\right), \tag{9}$$

with no leading constant. There is no ceiling: $p_{\text{cov}} \to 1$.

The two regimes therefore cross. Packing wins while the wasted overlap costs more than the gaps do, and covering wins once the gaps matter more. Equating (6) and (9) puts the crossover at

$$p^\star = 0.8028, \tag{10}$$

or $N \approx 84$ at $\sigma/r = 4.6$. Below $p^\star$ the packing law of (6) is the right tool and η_2 is a genuine bound. Above it, insisting on disjoint discs is simply a worse arrangement, and a design that needs $p = 0.95$ should abandon the separation constraint rather than conclude the target is unattainable.

This qualifies the headline of Section V rather than overturning it. The area-ratio audit of Table II stands: within the packing regime, that rule is optimistic and the $m = 2.5$ target is genuinely unreachable. But "no mesh exceeds 90.69%" is a statement about meshes constrained to $s \geq 2r$, and we state it that way. We note also that the separation constraint in our own configuration is well below $2r$, so the packing regime there reflects how the analytic policy chooses to arrange itself, not a limit imposed by the vehicles.

F. Which lever matters

Because $N \propto \sigma^2/r^2$, halving σ and doubling r are equally valuable, each reducing the required count fourfold, and no property of the placement algorithm changes the exponent. Placement quality determines how closely a team approaches this bound; it cannot move the bound.

TABLE III
MESH COST VS. SENSOR CADENCE ($\sigma_s = 4$ M, $r = 3$ M, REPOSITIONING WINDOW HELD AT 2.0 S SO THAT t_c IS CONSTANT)

fixes	rate (Hz)	R (m)	MC (m)	σ (m)	N at $p = 0.8$
3	0.90	11.68	11.66	8.26	29.4
4	1.34	11.00	10.96	7.78	26.1
5	1.79	10.34	10.47	7.31	23.0
6	2.24	9.75	9.76	6.89	20.5
8	3.14	8.79	8.71	6.22	16.7
12	4.93	7.47	7.43	5.28	12.0
16	6.72	6.60	6.57	4.67	9.4
24	10.31	5.50	5.50	3.89	6.5

Table III makes the point concretely on the sensing side, holding the repositioning window fixed so that t_c is constant and only the fix count varies. Raising the cadence from 3 to 24 fixes tightens σ by $2.1\times$ and reduces the mesh required for 80% coverage from 29 agents to 7. The competing lever behaves as the exponent dictates: at fixed cadence, moving r from 1 m to 8 m takes the requirement from 184 agents to 3.

VI. IS HEXAGONAL PACKING THE RIGHT ARRANGEMENT?

Equation (6) assumes a hexagonal arrangement, justified by Thue's theorem. That theorem concerns covering the infinite plane, whereas our mesh covers a finite Gaussian with a finite budget. The assumption therefore deserves an empirical check, and the check is informative about the relation between this problem and classical coverage control.

We compare five arrangements at matched N, σ and r: the hexagonal lattice used throughout; a Fermat-spiral (phyllotaxis) arrangement, which is locally near-hexagonal but free of grid artefacts; a square lattice at the same spacing, with packing

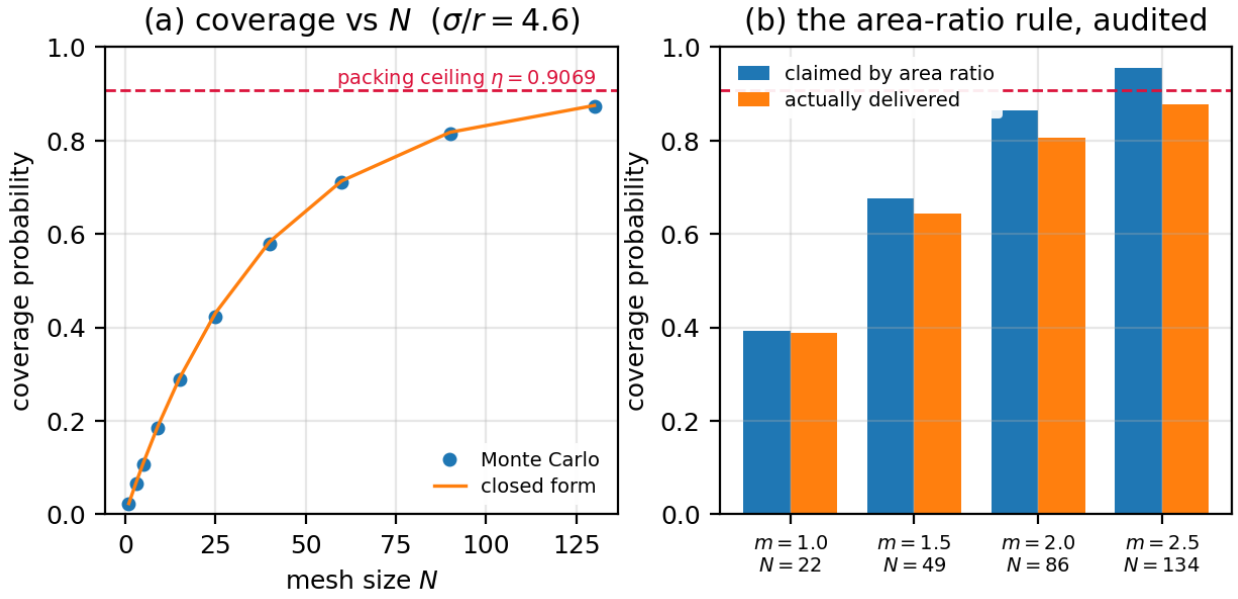


Fig. 2. (a) Coverage probability against mesh size: Monte Carlo against (6), with the packing ceiling η_2. (b) The area-ratio rule audited — claimed against delivered coverage at the N it prescribes.

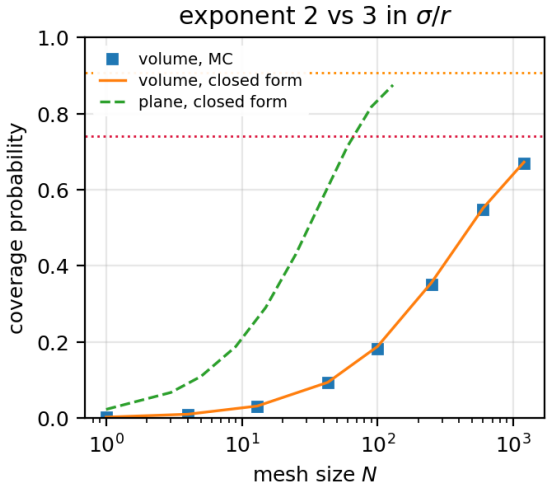


Fig. 3. Planar and volumetric coverage laws. Dotted lines mark the packing ceilings $\eta_2 = 0.9069$ and $\eta_3 = 0.7405$.

density $\pi/4$; a centroidal Voronoi configuration obtained by Lloyd's algorithm on samples from the density, which is what coverage control converges to [1], [2]; and uniform random placement subject to non-overlap, as a weak reference.

Table IV and Fig. 4 confirm the assumption: hexagonal packing is best at every budget tested, tied only by the square lattice at $N = 9$ where the arrangement is too small for the difference in density to express itself. The Fermat spiral trails it by a steady 5 to 7%, and the square lattice degrades as N grows, consistent with its lower packing density.

The centroidal Voronoi configuration is the interesting case.

TABLE IV
COVERAGE BY PLACEMENT STRATEGY AT MATCHED BUDGET ($\sigma = 9.25$ M, $r = 2$ M, 200 000 TRIALS PER CELL)

strategy	N=9	N=25	N=60	N=130
hexagonal lattice	**0.185**	**0.422**	**0.711**	**0.873**
Fermat spiral	0.171	0.401	0.676	0.831
square lattice	0.185	0.409	0.651	0.767
Lloyd / CVT	0.091	0.222	0.532	0.839
random, non-overlapping	0.153	0.327	0.520	0.655

At $N = 9$ it attains 0.495 of hexagonal coverage and at $N = 25$ only 0.524, converging to 0.962 by $N = 130$. The reason is not an implementation artefact but the objective: Lloyd's algorithm minimises mean squared distance to the nearest agent, which rewards placing agents out in the tails of the density where the squared-distance penalty is largest. Binary containment gives no credit for being closer to a point one does not cover. Measured nearest-neighbour spacing confirms it — the CVT solution rests at 5.34 m against the 4.0 m of a touching pack, spending budget on reach it cannot convert into coverage.

The practical reading is that the standard tool for this family of problems is the wrong tool for this member of it, and most wrong exactly where budgets are tight. It also sharpens the claim of Section X: the analytic baseline there is not merely a convenient heuristic but the best of five arrangements including the field's default.

VII. ROBUSTNESS OF THE ERROR MODEL

Equation (6) assumes an isotropic Gaussian error. Two departures are likely in practice. Heavy tails arise from occasional bad fixes, missed manoeuvres or track breaks.

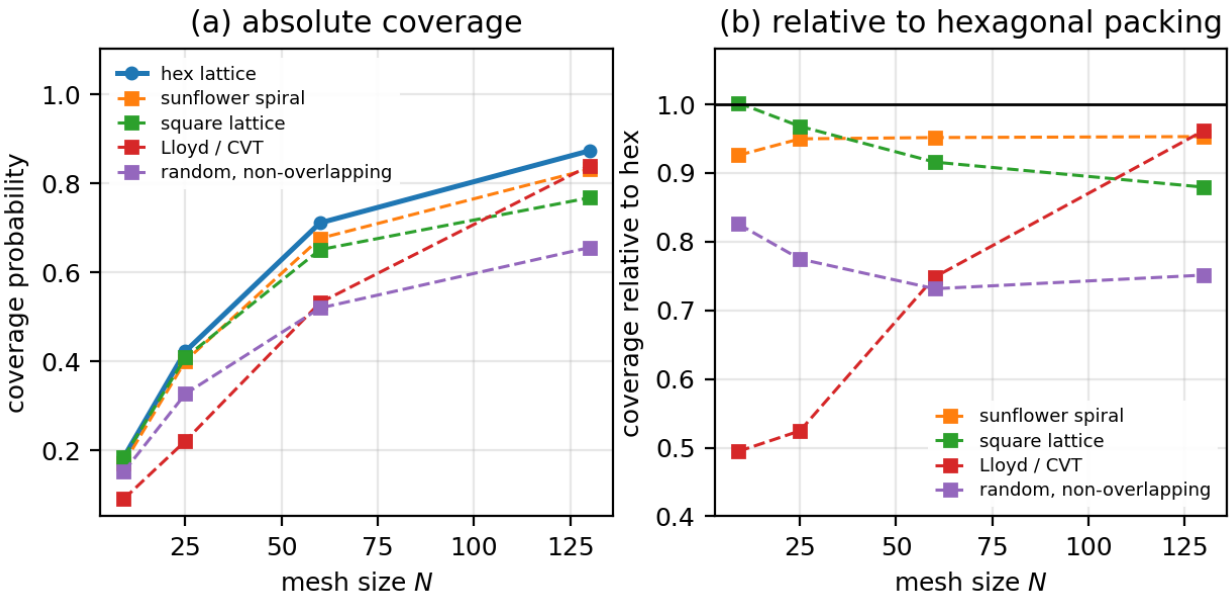


Fig. 4. Placement strategies at matched budget. (a) Absolute coverage. (b) Coverage relative to hexagonal packing. The centroidal Voronoi configuration is worst where the budget is tightest.

TABLE V
COVERAGE DELIVERED BY A MESH SIZED FOR A GAUSSIAN ($N = 43$, TARGET 0.60, $\sigma = 9.25$ M, $r = 2$ M)

error model	delivered	surplus	N for target
Gaussian (baseline)	0.602	+0.002	43
Student-t, $\nu = 10$	0.626	+0.026	39
Student-t, $\nu = 5$	0.663	+0.063	35
Student-t, $\nu = 3$	0.734	+0.134	24
elliptical 2:1	0.625	+0.025	40
elliptical 4:1	0.629	+0.029	37
elliptical 8:1	0.635	+0.035	37
biased 0.5σ	0.562	−0.038	47
biased 1.0σ	0.456	−0.144	67
biased 2.0σ	0.185	−0.415	136

Anisotropy arises because crossing geometry rarely leaves the two in-plane axes equally determined. A third, bias, arises when the predictor is systematically off-centre rather than merely imprecise — which is what an unmodelled manoeuvre produces.

We size a mesh *as if* the error were Gaussian, which is what a practitioner holding only σ would do, then measure what that mesh delivers against each true distribution. The normalisation of the shape departures matters and is stated explicitly: the Student-t cases hold the marginal variance at σ^2, and the elliptical cases hold the trace of the covariance at $2\sigma^2$, so that the RMS radial error matches the isotropic case and only the shape differs. The marginals cannot themselves be held fixed under anisotropy, since one axis must widen as the other narrows; under a determinant-preserving normalisation instead, the minor axis would contract less and the sign of the anisotropy result would change.

The result splits cleanly, and not in the direction we expected.

Shape departures make the law conservative. At fixed marginal σ, both heavy tails and anisotropy concentrate more mass near the mode than a Gaussian does, so a mesh sized by (7) over-delivers. A Student-t with $\nu = 3$ reaches the target with 24 agents where the Gaussian law prescribes 43. This is a statement about shape at fixed spread, and it does not license ignoring the error model: if the true σ exceeds the fitted one, the quadratic law applies to the true value and the cost is quadratic in the discrepancy.

Bias is the departure that breaks it. A one-sigma offset of the aim point costs 15 points of coverage, a two-sigma offset 42. Restoring the target takes 67 and 136 agents respectively, against 43 — a 3.2× inflation purchased purely to compensate for a mesh centred in the wrong place. This matters because Section VIII shows that unmodelled manoeuvre displaces the crossing point rather than blurring it, so it enters the budget here rather than as extra spread. The engineering reading is unambiguous: budget spent on the aim point dominates budget spent on team size, by a wide margin.

VIII. THE COMMIT-TIME WINDOW

A. The trade

Sections IV and V price a mesh given σ. A team that must physically reposition faces a further constraint, because σ is not exogenous: it depends on how early the team commits.

Let T be the commit time — the moment the team stops updating its aim point and flies to the formation. T is squeezed from both sides. Commit late and the prediction is sharp but there is no time to fly. Commit early and there is ample transit time but the arrival point must be extrapolated further.

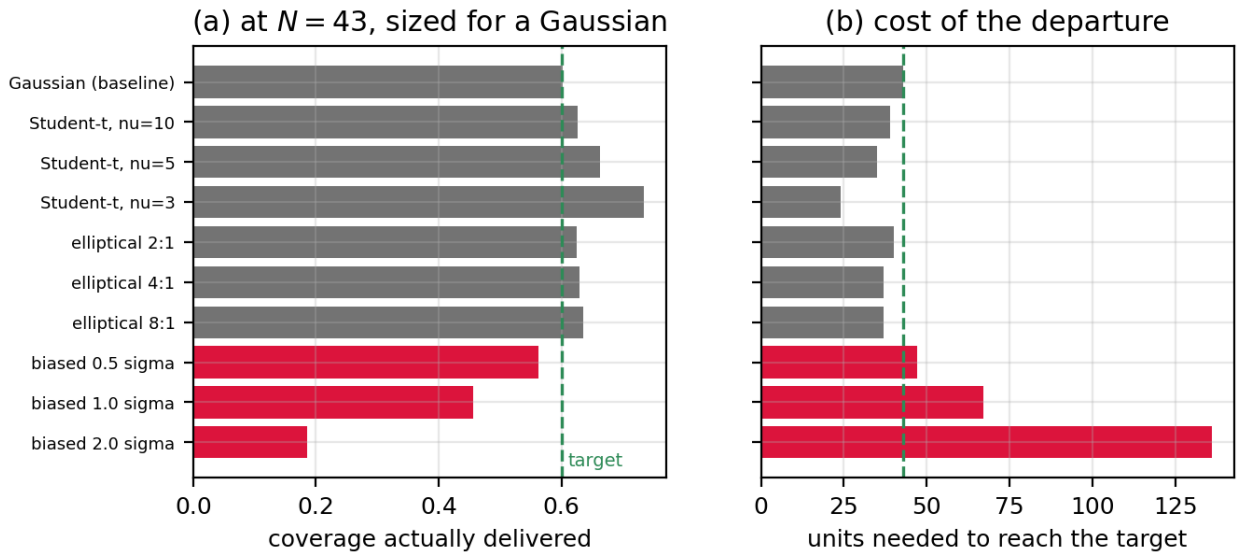


Fig. 5. Robustness to a misspecified error model. (a) Coverage delivered at a budget sized for a Gaussian. (b) Budget needed to restore the target. Shape departures (grey) leave the law conservative; bias (red) does not.

Two contributions enter $\sigma(T)$ in quadrature. The sensor term is (5), which *falls* as the track lengthens. The second is unmodelled lateral acceleration: the fit assumes a straight line, and an acceleration α displaces the crossing point by $\alpha T^2/2$, growing quadratically in the lead. We parameterise it by a per-axis standard deviation in units of g; this is a crude stand-in for the correlated-acceleration models of the tracking literature [13], [14], adequate because the conclusion turns on which term dominates rather than on its exact constant.

Including this second term is essential and its omission is not benign. In the configuration below — a track of roughly 30 s at 2.5 Hz — the sensor term alone gives $\sigma \approx 1$ m, which would imply a single agent suffices. Fig. 6a shows the actual budget: the sensor term stays between 0.9 and 2.5 m across the whole range of T, while the manoeuvre term runs from 2.5 m to 481 m. Two orders of magnitude separate them. The carefully derived sensor model of Section IV is, at these lead times, not the binding term.

B. An empty window

Feasibility requires that the worst assigned agent reach its slot. With a formation half-extent $L = r\sqrt{N/\eta_2}$ and a corridor of half-width C within which the formation centre may fall, the worst assigned transit is $\sqrt{2}(L + C)$, while an agent that must start and end at rest covers

$$d(T) = \begin{cases} aT^2/4, & T \le 2v_{\max}/a, \\ v_{\max}T - v_{\max}^2/a, & \text{otherwise.} \end{cases} \tag{11}$$

Note the two ramps: an agent that must arrive *and hold station* brakes as well as accelerates, so the familiar $v_{\max}T - v_{\max}^2/2a$ overstates reach by one ramp's worth.

For $C = 60$ m, $r = 6$ m, $v_{\max} = 33$ m/s, $a = 27$ m/s^2 and a manoeuvre uncertainty of $0.5g$, **no** T **satisfies both constraints**. The closest approach is at $T = 1.5$ s, short by 86 m of transit. The reason is visible in Fig. 6b: at small T the required transit is dominated by the corridor itself, $\sqrt{2}C \approx 85$ m, which the agent cannot cross in the time available regardless of how small the formation is.

Relaxing the transit model to an optimistic bound — assuming agents are already distributed around the formation centre, so that typical transit is C rather than $\sqrt{2}(L + C)$ — opens a window at $T \ge 3.5$ s. But the cheapest feasible mesh in that window costs 97 agents. The static alternative fares no better. Covering a fraction p of a corridor of radius C with disjoint discs needs $N = pC^2/r^2$, here 80 agents, but the corridor cannot hold them: walking the lattice outward until a disc would cross the boundary admits only 73, capping static coverage at 0.730. At the stated target of 0.80 this configuration therefore admits no solution at all — neither repositioning nor blanketing.

Table VI sweeps corridor width against manoeuvre uncertainty. A window exists only in the left-hand column, at $0.1g$ — that is, only if the tracked point flies very nearly straight. At $0.5g$ and above the window is empty at every corridor width tested.

Stated in the terms of the application, this is the sharpest engineering result in the paper. A mesh that repositions onto a prediction can defend a site against an object that flies nearly straight, and cannot defend it against one that manoeuvres: the boundary sits between $0.1g$ and $0.5g$ at every corridor width we tested. The limit belongs to the geometry rather than to the vehicles. At small commit times the transit an agent must fly is dominated by the corridor itself, $\sqrt{2}C \approx 85$ m, which is already beyond reach before the formation's own extent is counted; closing the 86 m shortfall at the best commit time found would take an acceleration near $18g$. Faster or more

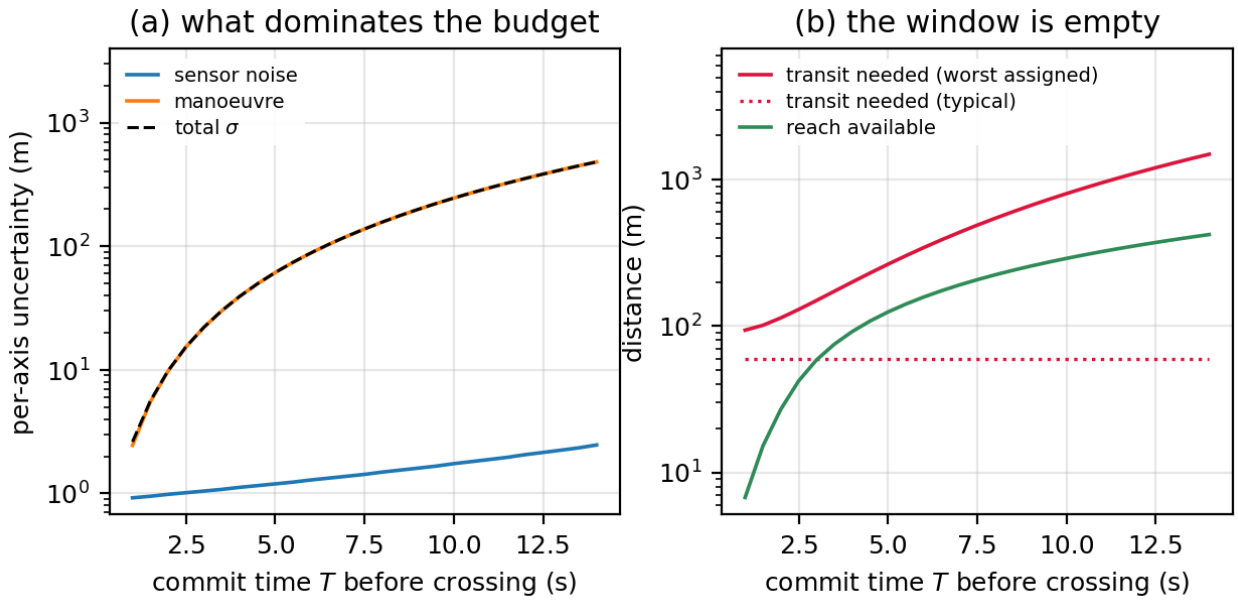


Fig. 6. (a) The uncertainty budget against commit time: the sensor term is negligible beside unmodelled manoeuvre at every T. (b) Transit required against reach available; the two bounds on required transit bracket the answer, and neither admits a useful window.

TABLE VI
CHEAPEST FEASIBLE MESH SIZE; "—" WHERE THE COMMIT WINDOW IS EMPTY. COUNTS ARE CONTINUOUS, PER SECTION III

corridor C	$0.1g$	$0.5g$	$1.0g$	$3.0g$
±15 m	0.8	—	—	—
±25 m	1.6	—	—	—
±40 m	4.0	—	—	—
±60 m	16.4	—	—	—
±90 m	93.7	—	—	—

agile interceptors therefore do not open the window at any plausible margin. A narrower corridor, an earlier and better prediction, or a larger engagement radius do — which is where the design effort belongs.

IX. APPLYING THE BOUNDS

The results above are individually simple, and their value is in the order they are applied; we set out that order below.

Step 1: price the prediction, including what the fit does not model. Compute R from (5) for the intended cadence and lead time, then add the manoeuvre term of Section VIII in quadrature. Section VIII is emphatic about the order of magnitude involved: a long track drives the sensor term to a metre or two while the manoeuvre term reaches hundreds, so a budget containing only the first is not conservative but simply wrong. If the tracked point may accelerate at all, the manoeuvre term is the budget.

Step 2: convert the budget into a team size. Apply (7), not an area ratio. Check the target against the ceiling first: while $s \geq 2r$, any $p \geq \eta_2$ is unattainable at every N (if the platform tolerates $s < 2r$, use (9) instead and there is no ceiling), and an area ratio will nonetheless return a finite number. If the arrival altitude is also free, use the volumetric form and expect roughly an order of magnitude more agents.

Step 3: check that the team can physically be where the law puts it. The count from Step 2 implies a formation half-extent $L = r\sqrt{N/\eta_2}$, which with the corridor half-width sets a transit distance. Compare it against (11) at the intended commit time. If no commit time satisfies both constraints, no placement policy can succeed and the configuration must change — which is a conclusion available in seconds, before any controller exists.

Step 4: compare against blanketing. Repositioning is not automatically worthwhile. A static mesh covering a fraction p of the corridor costs pC^2/r^2 agents and needs no prediction, no commit decision and no transit. Check it against capacity as well as against cost: the corridor holds at most $\eta_2 C^2/r^2$ discs asymptotically and fewer once the boundary is accounted for, so a target above that capacity is unreachable statically however many agents are available. In Section VIII's configuration it is cheaper in nominal count (80 against 97) yet fails the capacity check, which is exactly why the check belongs in this step: neither option is feasible at the stated target. The comparison takes two lines and settles whether the repositioning concept is worth building at all.

Step 5: decide where to spend. Because $N \propto \sigma^2/r^2$, halving σ and doubling r are worth a factor of four each, and no placement algorithm changes that exponent. Section VII adds a sharper version: bias in the aim point costs more than spread. A one-sigma offset demands $1.6\times$ the agents and a two-sigma offset $3.2\times$, so effort spent on removing a systematic aiming error dominates effort spent on fleet size, and dominates it by more than the quadratic law alone suggests.

Step 6: only now consider learning. Steps 1–5 produce a feasibility ceiling, a null-policy score and a classical baseline. These are the three numbers that make a learned result interpretable, and Section VI supplies a fourth consideration: the classical baseline should be a packed arrangement, since the centroidal Voronoi configuration that coverage control would naturally supply is the wrong objective for binary containment and loses half its coverage at tight budgets.

Applied in this order, the analysis of a new configuration costs minutes. The sequence also makes clear which failures are worth investigating: a policy scoring near the ceiling is not underperforming, and a policy scoring near the null score on a task whose ceiling is high is a genuine finding about the learner rather than about the task.

X. Implications for Learned Placement

A. Measured outcomes

The bounds above were derived after, not before, a sequence of reinforcement-learning experiments on this task in an accompanying prototype. We report those outcomes because the sequence itself is the argument.

The geometry is the corrected one of Section X-C: 25 agents, sector ± 35 m, corridor ± 20 m, $r = 3$ m, mean $\sigma = 4.57$ m, a 4 s window. All policies are evaluated through the identical environment code path over the same 100 episode seeds, with percentile bootstrap intervals [33] over episodes.

Before the comparison, the denominator — and here the analysis of Section V-E has to be applied to our own configuration rather than left as a general remark.

A crossing falls inside the reachable sector in every episode, so reachability contributes a factor of 1.000. For the coverage factor, the environment's separation limit is $s = 2.5$ m against $2r = 6$ m. By Section V-E the packing law therefore does *not* apply: discs may overlap, and the governing constant is the covering density. The two laws give 0.904 by (6) and 0.988 by (9) for 25 agents at $\sigma = 4.57$ m and $r = 3$ m, so the applicable ceiling is $1.000 \times 0.988 = 0.988$. Table VII reports both columns.

Two consequences follow. Quoting the packing ceiling would have flattered every policy by roughly eight percentage points. And the packing ceiling of 0.904 sits *above* the crossover $p^\star = 0.8028$ of (10), so by this paper's own analysis the hexagonal arrangement is not optimal at this operating point: it is the best of the five *packed* arrangements of Section VI, and it is what the analytic policy implements, but an overlapping arrangement would do better. We did not notice this until the covering regime was worked out.

The learner was PPO [21] as implemented in Stable-Baselines3 [23], with a two-layer 128×128 MLP policy, learning rate 3×10^{-4}, 2048 steps per rollout, minibatch 256, $\gamma = 0.995$, trained for 4×10^5 environment steps at each seed. No hyperparameter search was run.

Because every policy is evaluated on the same episodes, the episode is a matched unit and the comparison against the null policy should be *paired*. Table VIII reports the bootstrap of the per-episode difference. At all three trainer seeds the interval

TABLE VII
Placement Policies Over 100 Episode Seeds Through an Identical Environment Path. Intervals Are 95% Percentile Bootstrap Over Episodes; PPO Is Reported at Three Trainer Seeds. The Covering Column Is the Applicable One, Since $s < 2r$

			of ceiling	
policy	hit rate	coll.	packing	covering
null (static lattice)	0.14 [0.08, 0.21]	**0.00**	15.5%	14.2%
centralised PPO, seed 0	0.12 [0.06, 0.19]	0.78	13.3%	12.1%
centralised PPO, seed 1	0.19 [0.12, 0.27]	0.74	21.0%	19.2%
centralised PPO, seed 2	0.14 [0.08, 0.21]	0.61	15.5%	14.2%
mean over seeds	0.15	0.71	16.6%	15.2%
analytic hex packing	**0.84** [0.77, 0.91]	0.62	92.9%	**85.0%**
+ transit repulsion	0.79 [0.71, 0.87]	0.49	87.3%	79.9%

TABLE VIII
Paired Bootstrap of the Per-Episode Difference Against the Null Policy, Same 100 Episodes

policy	mean difference	95% CI
centralised PPO, seed 0	−0.020	[−0.110, +0.070]
centralised PPO, seed 1	+0.050	[−0.060, +0.160]
centralised PPO, seed 2	+0.000	[−0.100, +0.100]
analytic hex packing	**+0.700**	[+0.600, +0.790]
+ transit repulsion	+0.650	[+0.550, +0.750]

contains zero; for both analytic policies it does not, by a wide margin. This is the statement the paper relies on. Pairing is the comparison a matched design licenses, not automatically a tighter one. Here the episode-level correlation between each policy and the null policy runs from -0.12 to $+0.12$, so the paired intervals are in fact slightly *wider* than the marginals of Table VII. Pairing buys correctness rather than precision in this instance; arguing from overlapping independent intervals would not have been valid.

Averaged over trainer seeds the learned policy scores 0.15 against the null policy's 0.14, with a seed-to-seed spread from 0.12 to 0.19 that straddles the null policy in both directions. On a single seed the conclusion would have been either "PPO is worse than doing nothing" or "PPO beats doing nothing by a third", depending which seed was drawn; this is the failure mode [31] quantifies.

The analytic solution reaches 85.0% of the applicable ceiling. By Section VI, hexagonal packing is the best of the five *packed* arrangements at this budget, so the residual 15% is not all headroom for a smarter policy of the same kind: part is distance to the bound, and part is the packing–covering gap identified above, which no packed arrangement can close. PPO's explained variance never left the neighbourhood of zero, ending at -0.12: the value function did not fit the task at all.

PPO's collision rate of 61–78% is high, but so is the analytic policy's 62%, and the repulsion variant only brings it to 49%. Collisions are therefore *not* a discriminator between learning and the analytic method here; both incur them and only the null policy, which never moves, avoids them entirely. They arise in transit, when agents converging on a formation from

opposite sides of the sector cross paths, rather than in the settled formation — which is why a short-range repulsion term reduces them at a small cost in coverage. A deployment would need to address this for either method, and we do not claim the analytic solution is collision-safe.

An initial "retrain" at the released configuration reproduced the original run to eight decimal places in every training statistic, because the trainer seed is pinned in the configuration file. It is a determinism check and nothing more; it is reported separately in the released records and excluded from the mean above, since counting it would fabricate a replicate.

B. Why this is unsurprising

The placement problem is not an unstructured control problem. It decomposes into two classical sub-problems with known solutions: where the formation should sit, which is densest packing over a radially symmetric density; and which agent occupies which slot, which is linear assignment [16]. A monolithic policy is asked to rediscover both in a 50-dimensional continuous action space whose parameterisation treats interchangeable agents as unrelated output dimensions — precisely the symmetry that makes the assignment sub-problem tractable, and precisely what permutation-equivariant architectures [24]–[26] exist to supply.

Concentration is overwhelmingly the right strategy: 84% against 14% for holding station. What fails is learning it end to end.

C. Three faults that mimic learning failure

Before the comparison above was meaningful, three specification faults had to be removed. Each produced a symptom easily misread as "the policy did not learn". They are instances of a general phenomenon [27], [28], but the specific forms are worth recording.

An infeasible scenario. The first configuration had a ceiling near 10%, decomposed as 56.2% for the arrival falling inside the reachable area times 18.1% for the best-case coverage of nine agents at $r = 2$ m and $\sigma = 9.25$ m. Five rounds of reward shaping were spent on a policy already operating at or above what the configuration permitted; measured hit rates of 4–16% were between 40% and 160% of the ceiling. The correct first diagnostic costs seconds and was performed only after the fact.

Penalty terms of the wrong order in N. Pairwise separation violations were charged per unordered pair, which grows as $O(N^2)$, while the coverage reward was a mean over agents, which is $O(1)$. At $N = 25$ the penalty exceeded the entire achievable coverage reward by more than an order of magnitude, making dispersal reward-optimal. The diagnostic signature was explained variance pinned near 10^{-7} alongside a large value loss. Every term in a multi-agent reward must carry the same order in N.

A baseline that already solves the task. After the previous fix, the mesh was dense enough that the resting lattice scored 63% with no control input at all, making any learned result unfalsifiable. The corrected geometry holds a wide sector sparsely, at a lattice spacing far above r, so that the null policy scores 14%.

D. The recommendation

For constrained multi-agent coverage tasks we suggest three things be computed and reported before any learning experiment, and quoted alongside every result:

1) the feasibility ceiling, as the product of reachability and best-case coverage;
2) the null-policy score, so that a task the null policy already solves is caught before training rather than after;
3) a closed-form or classical-algorithm baseline, evaluated through the same code path as the learner.

A bare success rate without these is uninterpretable. In this project, three of the four things that looked like learning failures were scenario or reward specification faults, and the fourth — the genuine one — is explained by Section VIII without reference to the learner at all.

XI. Limitations

The manoeuvre model is crude. Unmodelled acceleration enters as an independent per-axis Gaussian with fixed standard deviation, integrated over the lead time. A real manoeuvring point is correlated in time and bounded in magnitude [13], [14], and a constant-acceleration fit would absorb part of what we charge to uncertainty. The qualitative conclusion — that this term dominates beyond a few seconds of lead — is robust to the details; the specific numbers in Table VI are not.

The transit bounds bracket rather than resolve. We report a worst-assignment bound and an optimistic bound because the true requirement depends on the resting layout, which we do not model. The conclusion that repositioning loses holds under both, but a layout-aware analysis would give a single number.

Point-mass agents. Attitude dynamics, actuator lag and downwash interactions are absent. In a separate full-physics run of the accompanying prototype the formation settled 10–27 m from its slots against the 10–15 m predicted here, so the planar model is optimistic but not by an order of magnitude.

Centralised control. One policy emits all N commands, and assignment is computed globally. Decentralised operation on local information is the natural next step; the assignment sub-problem admits distributed solutions [17]–[19], but the packing sub-problem would need a consensus mechanism we have not designed.

The negative learning result concerns one architecture. An unstructured centralised policy loses to analytic packing across three seeds. A permutation-equivariant policy with assignment supplied by the Hungarian algorithm and only residual corrections learned has not been tried, and until it has, the result is about end-to-end learning rather than about learned placement in general. Given how strongly Section VI favours the packed arrangement, we would expect a structured learner to approach it rather than beat it.

Single prediction. The analysis computes one prediction and commits. Continuous re-prediction with a shrinking σ is the obvious extension and would change the commit-time analysis qualitatively, since the team could re-converge as the estimate sharpens.

XII. Conclusion

We have given closed forms, each validated numerically, for the quantities that bound multi-agent coverage of an uncertain arrival point before any policy is chosen: the propagated uncertainty of the prediction, accurate to 1.3% against Monte Carlo and calibrated to within a point of nominal; the coverage probability of a packed mesh in the plane and in a volume, accurate to 0.007 and 0.005 absolute, whose quadratic and cubic exponents are exact by construction and whose counts agree with Monte Carlo inversion to within 16%; and the admissible window of commit times, which can be empty.

Four consequences follow. There is a ceiling on coverage — 90.69% in the plane, 74.05% in a volume — that no arrangement of mutually separated agents can exceed at any size, and an area-ratio rule that misses it prescribes finite team sizes for unattainable targets. That ceiling belongs to the separation constraint rather than to the problem: permit overlap and the governing constant becomes the thinnest-covering density [8], above a crossover at $p = 0.80$ this is the better arrangement, and no ceiling remains — so a design needing high coverage should relax the constraint rather than declare the target impossible. The arrangement also matters within the packing regime, and the standard tool is the wrong one: a centroidal Voronoi configuration attains half the coverage of a packed one at tight budgets, because it optimises distance rather than containment. And the Gaussian assumption is conservative under heavy tails and anisotropy but fragile under bias, which is exactly the form unmodelled manoeuvre takes.

In the terms of the application that motivated the work, the answer is negative and specific. A repositioning mesh of interceptors can defend a site against a nearly ballistic object and cannot defend it against one manoeuvring at half a g or more; the binding constraint is the transit imposed by the corridor width, not the agility of the vehicles, and no placement policy, learned or analytic, can lift it.

The methodological claim is the one we would most like to see adopted. A success rate reported for a hand-designed multi-agent task, without the feasibility ceiling, the null-policy score and a classical baseline alongside it, does not carry enough information to be interpreted. In the case documented here it led to five rounds of optimisation aimed at a component that was never the binding constraint — and, had we stopped at one trainer seed, to a confident conclusion in whichever direction that seed happened to point.

Reproducibility

The prototype this work analyses — the prediction module, the hexagonal lattice generator, the placement environment and the training scripts — is openly available at https://github.com/DmytroBabarytskyi/drone-training, at commit `4b42752`. The eight experiment scripts, the figure generator and the JSON record of every run reported here are released alongside it,[1] together with the training configurations for all three trainer seeds and the corresponding checkpoints.

Three properties of the release matter. Each experiment script imports the prediction, lattice and environment code from the prototype rather than reimplementing its formulas, so a validation run tests the implementation that actually runs. Every Monte Carlo seed is fixed and stated in the sources. And every policy in Table VII is evaluated over the same episode seeds through the same environment code path, so the comparison is like-for-like rather than a comparison of evaluation harnesses. A single entry point, `run_all.py`, regenerates every number and every figure in this paper in roughly six minutes on a laptop CPU.

[1] See the repository's `README` for the exact location of the experiment scripts and the archived policy checkpoints.

Declaration on Generative AI

During the preparation of this work, the author used Claude (Anthropic) in order to: search and screen the literature; write the code that produces Figures 1–5 from the recorded results; paraphrase and format the manuscript.

The research programme and its framing, the prototype and simulation environment that every experiment runs against, the reinforcement-learning study whose outcomes Section X reports are the author's own. Every closed form in this paper is independently checked against Monte Carlo simulation, and those checks are reported in full. After using this tool, the author reviewed and edited the content as needed, and takes full responsibility for the publication's content.

References

[1] J. Cortés, S. Martínez, T. Karatas, and F. Bullo, "Coverage control for mobile sensing networks," *IEEE Trans. Robot. Autom.*, vol. 20, no. 2, pp. 243–255, 2004.

[2] Q. Du, V. Faber, and M. Gunzburger, "Centroidal Voronoi tessellations: applications and algorithms," *SIAM Review*, vol. 41, no. 4, pp. 637–676, 1999.

[3] M. Schwager, D. Rus, and J.-J. Slotine, "Decentralized, adaptive coverage control for networked robots," *Int. J. Robotics Research*, vol. 28, no. 3, pp. 357–375, 2009.

[4] F. Bullo, J. Cortés, and S. Martínez, *Distributed Control of Robotic Networks*. Princeton, NJ, USA: Princeton University Press, 2009.

[5] W. Gosrich, S. Mayya, R. Li, J. Paulos, M. Yim, A. Ribeiro, and V. Kumar, "Coverage control in multi-robot systems via graph neural networks," in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, 2022, pp. 8787–8793.

[6] L. Fejes Tóth, *Lagerungen in der Ebene, auf der Kugel und im Raum*. Berlin, Germany: Springer, 1953.

[7] J. H. Conway and N. J. A. Sloane, *Sphere Packings, Lattices and Groups*, 3rd ed. New York, NY, USA: Springer, 1999.

[8] R. Kershner, "The number of circles covering a set," *American Journal of Mathematics*, vol. 61, no. 3, pp. 665–671, 1939.

[9] T. C. Hales, "A proof of the Kepler conjecture," *Annals of Mathematics*, vol. 162, no. 3, pp. 1065–1185, 2005.

[10] T. C. Hales *et al.*, "A formal proof of the Kepler conjecture," *Forum of Mathematics, Pi*, vol. 5, e2, 2017.

[11] R. E. Kalman, "A new approach to linear filtering and prediction problems," *Trans. ASME J. Basic Engineering*, vol. 82, no. 1, pp. 35–45, 1960.

[12] Y. Bar-Shalom, X.-R. Li, and T. Kirubarajan, *Estimation with Applications to Tracking and Navigation*. New York, NY, USA: Wiley, 2001.

[13] R. A. Singer, "Estimating optimal tracking filter performance for manned maneuvering targets," *IEEE Trans. Aerospace and Electronic Systems*, vol. AES-6, no. 4, pp. 473–483, 1970.

[14] X. R. Li and V. P. Jilkov, "Survey of maneuvering target tracking. Part I: dynamic models," *IEEE Trans. Aerospace and Electronic Systems*, vol. 39, no. 4, pp. 1333–1364, 2003.

[15] G. W. Oehlert, "A note on the delta method," *The American Statistician*, vol. 46, no. 1, pp. 27–29, 1992.

[16] H. W. Kuhn, "The Hungarian method for the assignment problem," *Naval Research Logistics Quarterly*, vol. 2, no. 1–2, pp. 83–97, 1955.

[17] D. P. Bertsekas, "The auction algorithm: a distributed relaxation method for the assignment problem," *Annals of Operations Research*, vol. 14, no. 1, pp. 105–123, 1988.

[18] M. M. Zavlanos, L. Spesivtsev, and G. J. Pappas, "A distributed auction algorithm for the assignment problem," in *Proc. 47th IEEE Conf. Decision and Control*, 2008.

[19] H.-L. Choi, L. Brunet, and J. P. How, "Consensus-based decentralized auctions for robust task allocation," *IEEE Trans. Robotics*, vol. 25, no. 4, pp. 912–926, 2009.

[20] T. H. Chung, G. A. Hollinger, and V. Isler, "Search and pursuit-evasion in mobile robotics: a survey," *Autonomous Robots*, vol. 31, no. 4, pp. 299–316, 2011.

[21] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," arXiv:1707.06347, 2017.

[22] C. Yu, A. Velu, E. Vinitsky, J. Gao, Y. Wang, A. Bayen, and Y. Wu, "The surprising effectiveness of PPO in cooperative multi-agent games," in *Advances in Neural Information Processing Systems 35, Datasets and Benchmarks Track*, 2022.

[23] A. Raffin, A. Hill, A. Gleave, A. Kanervisto, M. Ernestus, and N. Dormann, "Stable-Baselines3: reliable reinforcement learning implementations," *J. Machine Learning Research*, vol. 22, no. 268, pp. 1–8, 2021.

[24] P. W. Battaglia *et al.*, "Relational inductive biases, deep learning, and graph networks," arXiv:1806.01261, 2018.

[25] I.-J. Liu, R. A. Yeh, and A. G. Schwing, "PIC: permutation invariant critic for multi-agent deep reinforcement learning," in *Proc. Conf. Robot Learning (CoRL)*, PMLR vol. 100, 2020, pp. 590–602.

[26] E. van der Pol, H. van Hoof, F. A. Oliehoek, and M. Welling, "Multi-agent MDP homomorphic networks," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2022.

[27] A. Y. Ng, D. Harada, and S. Russell, "Policy invariance under reward transformations: theory and application to reward shaping," in *Proc. ICML*, 1999, pp. 278–287.

[28] J. Skalse, N. H. R. Howe, D. Krasheninnikov, and D. Krueger, "Defining and characterizing reward hacking," in *Advances in Neural Information Processing Systems 35*, 2022.

[29] R. S. Sutton and A. G. Barto, *Reinforcement Learning: An Introduction*, 2nd ed. Cambridge, MA, USA: MIT Press, 2018.

[30] P. Henderson, R. Islam, P. Bachman, J. Pineau, D. Precup, and D. Meger, "Deep reinforcement learning that matters," in *Proc. AAAI*, 2018, pp. 3207–3214.

[31] C. Colas, O. Sigaud, and P.-Y. Oudeyer, "How many random seeds? Statistical power analysis in deep reinforcement learning experiments," arXiv:1806.08295, 2018.

[32] R. Agarwal, M. Schwarzer, P. S. Castro, A. Courville, and M. G. Bellemare, "Deep reinforcement learning at the edge of the statistical precipice," in *Advances in Neural Information Processing Systems 34*, 2021.

[33] B. Efron and R. J. Tibshirani, *An Introduction to the Bootstrap*. New York, NY, USA: Chapman & Hall, 1993.

[34] C. R. Harris *et al.*, "Array programming with NumPy," *Nature*, vol. 585, pp. 357–362, 2020.

[35] P. Virtanen *et al.*, "SciPy 1.0: fundamental algorithms for scientific computing in Python," *Nature Methods*, vol. 17, pp. 261–272, 2020.